

Writing Is Coding: When Writing Becomes the Starting Point for AI Cross-Media Learning and Narrative

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ABSTRACT

Writing in the digital age has transformed from a mere communicative practice into a computational interface for multimodal creation—establishing a new paradigm where natural language functions as executable code for artificial intelligence systems. This research examines the pedagogical implications of this “Writing is Coding” framework, focusing on how learners navigate cross-media AI co-creation processes. Through a comprehensive three-round experiment, 102 undergraduate students engaged in progressive AI collaboration across textual, visual, and video modalities, using natural language as their primary creative instrument. Employing grounded theory methodology, the researchers analyzed data from student reflection journals, teacher observation logs, and semi-structured interviews to identify emergent patterns in AI-human creative partnerships. Five interconnected themes emerged: (1) prompt engineering as algorithmic thinking, (2) multimodal literacy development, (3) collaborative negotiation of creative agency, (4) emotional engagement through iterative refinement, and (5) metacognitive awareness of human-AI boundaries. Findings suggest that the “Writing is Coding” framework facilitates a deeper understanding of computational processes while enhancing creative expression. The three-round progression demonstrated increasing student comfort with AI collaboration, though tensions around authorship and creative identity persisted. This research contributes to emerging pedagogical approaches that position writing as a form of programming in AI-enhanced learning environments, with implications for curriculum design and digital literacy development.

KEYWORDS:

AI Co-Creation, Writing is Coding, Multimodal Learning, Prompt Engineering, Digital Literacy, Generative AI, Educational Technology

1. INTRODUCTION

In today's digital landscape, the relationship between human language and computational processes has undergone a profound transformation. The emergence of generative artificial intelligence (AI) has redefined writing as more than mere expression—it has become a form of programming, where natural language functions as code that instructs AI systems to produce diverse outputs across modalities. This paradigm shift, which we conceptualize as “Writing is Coding,” represents a fundamental reconceptualization of literacy in the age of AI co-creation (DePalma & Alexander, 2018; Smith, 2019).

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Traditional approaches to AI interaction have often emphasized directive control through specialized programming languages or structured commands. However, the evolution of large language models and multimodal generative systems has enabled a more natural, conversational approach to human-AI collaboration. Rather than programming AI through explicit code, users can now engage in co-creative processes through natural language prompts—effectively “coding” with their words to produce sophisticated outputs in text, image, and video formats (Tan et al., 2024; Tedre & Vartiainen, 2023).

The emergence of generative AI tools like MidJourney and DALL-E has democratized access to complex creative processes by enabling users to generate sophisticated visual outputs with minimal technical expertise. These tools align with the low-floor/high-ceiling principle in education, allowing beginners to engage while offering advanced users opportunities for refinement. However, this democratization raises questions about authorship and the authenticity of style in AI-assisted works (Frosio, 2025).

In educational contexts, multimodal composition has gained traction as a pedagogical approach for fostering creativity and collaboration (Scanlon, 2015; Jocius, 2018). Distributed collaboration frameworks highlight how students toggle between coding prompts and curating outputs in group projects. For example, adolescents composing across digital projects often adopt alternating lead roles or balanced divisions of labor. These collaborative dynamics mirror the recursive nature of working with generative AI tools: students must iteratively refine their prompts to align outputs with their rhetorical intentions (DePalma & Alexander, 2018; Smith, 2019).

Despite its transformative potential, generative AI also presents challenges. Scholars have noted that while AI enhances individual storytelling quality by reducing technical barriers (Ali et al., 2024), it risks homogenizing collective creativity due to algorithmic biases (Mukherjee & Chang, 2023). This “social dilemma” underscores the tension between empowering creators and preserving diversity in creative outputs (Garcia et al., 2024). Furthermore, insufficient prompt literacy—the ability to effectively craft inputs for desired outputs—can hinder students' ability to fully realize their creative visions.

This study investigates how students navigate this emerging landscape of AI co-creation across multiple modalities. The researchers posit that understanding writing as a form of coding offers valuable insights into both computational thinking and creative expression in educational contexts. By examining how students engage with AI systems through natural language rather than directive control, we aim to illuminate the cognitive, creative, and pedagogical dimensions of this new literacy paradigm.

The research addresses the following three research questions:

1. How do students conceptualize and adapt their writing processes when engaging in AI co-creation across text, image, and video modalities?
2. What cognitive and creative strategies emerge as students progress through multiple rounds of increasingly complex multimodal AI collaboration?
3. How does the “Writing is Coding” framework influence students' understanding of both computational processes and creative expression?

To explore these questions, we designed a three-round experimental study involving 102 undergraduate students from diverse academic disciplines. Each round focused on a different modality: text generation in Round 1, image creation in Round 2, and video production in Round 3. This progressive approach allowed us to observe how students' engagement with AI co-creation evolved as they moved from familiar textual domains to more complex multimodal environments.

The significance of this research lies in its potential to inform educational approaches that integrate AI co-creation into curriculum design. As generative AI becomes increasingly accessible in educational settings, understanding how students conceptualize and engage with these tools becomes

crucial for developing effective pedagogical frameworks. The “Writing is Coding” paradigm offers a conceptual bridge between computational thinking and creative expression—two domains often treated as separate in traditional educational models.

Furthermore, this study contributes to ongoing discussions about digital literacy in an AI-augmented world. As the boundaries between human and machine creativity become increasingly fluid, students must develop not only technical skills but also critical awareness of how their language shapes computational outputs. By examining the strategies, challenges, and insights that emerge through multimodal AI co-creation, this research aims to advance our understanding of what it means to be literate in an age where writing functions as a form of programming (Hsu et al., 2018; Thornberg & Dunne, 2019).

In the following sections, we review relevant literature on AI co-creation, multimodal learning, computational thinking in education, and grounded theory methodology. The researchers then detail our methodological approach, present findings from each round of the experiment, and discuss the theoretical and practical implications of the “Writing is Coding” framework for educational practice and research.

2. LITERATURE REVIEW

2.1 Natural Language as Computational Interface

The conceptualization of writing as a computational interface has evolved significantly with the advent of generative AI technologies. Traditionally, computational writing referred primarily to the use of digital tools to enhance or facilitate conventional writing processes (Warschauer, 2010). However, recent scholarship has begun to explore the deeper parallels between natural language composition and computational thinking. Vee (2017) argues that literacy has always involved a form of algorithmic thinking, as writers must consider the logical structure, sequential organization, and conditional relationships within their texts. This perspective has gained new relevance as AI systems increasingly respond to natural language as a form of executable instruction.

In the context of generative AI, writing transcends its traditional boundaries as a linguistic endeavor and emerges as a form of computational multimodality - a practice that unites expressive language, algorithmic thinking, and multimodal output generation. This perspective redefines text not only as narrative but also as executable code capable of orchestrating color, motion, sound, and spatial aesthetics through AI mediation (DePalma & Alexander, 2018; Macleroy, 2016).

This convergence is particularly salient in poetic composition, where metaphoric language and emotional nuance serve as potent inputs for multimodal synthesis. When students employ poetic prompts to generate visuals or cinematic expressions, they engage in a unique form of computational authorship. They debug lines of verse much like lines of code—revising tone, rhythm, or metaphor to produce intended visual or emotional effects (Jocius, 2018; Tedre & Vartiainen, 2023). This iterative process highlights the alignment between creative writing and algorithmic logic, emphasizing abstraction, pattern recognition, and system feedback as integral to both practices (Kalelioglu et al., 2016; Sanford & Naidu, 2016).

Lin et al. (2012), in their study of affective interfaces for a digital arts tutoring system, offer a prescient example of how emotion-driven text inputs can guide visual interpretation. Their system utilized OMCSNet and WordNet to infer emotional states from textual cues and translated these into dynamic visual agent expressions. This early model of affective multimodality revealed that poetic text, when paired with an AI system responsive to emotional valence, could shape user experience and enhance learner engagement. The feedback loop between written expression and affective response underscores how computational multimodality can scaffold both emotional articulation and visual creativity.

The “Writing is Coding” paradigm builds upon these insights by explicitly framing natural language as a form of computational instruction in AI co-creation. Rather than viewing AI tools as

mere assistants that augment traditional writing processes, this framework positions writing itself as a form of programming that directs AI systems across multiple modalities. This perspective offers valuable opportunities for integrating computational thinking into humanities education without requiring students to learn conventional programming languages.

2.2 AI Co-Creation in Educational Contexts

The integration of AI co-creation tools in educational settings has generated significant research interest in recent years. Studies have examined how these technologies influence student engagement, creativity, and learning outcomes across various disciplines. Generative AI simultaneously expands and constrains creative potential, producing what scholars term a “social dilemma of creativity” (Garcia et al., 2024; Mukherjee & Chang, 2023). On one hand, AI tools such as MidJourney, DALL-E, and GPT-4 enhance individual storytelling by lowering technical barriers and accelerating the production of sophisticated multimodal outputs. On the other hand, this acceleration often comes with a trade-off: the homogenization of stylistic voice across creators due to underlying algorithmic biases. Zhai and Nehm (2023) found a 22–26% improvement in story enjoyability when AI was used collaboratively; however, they also noted a convergence of style across groups, suggesting that while the AI elevates technical quality, it may erode the diversity of expressive nuance.

Research by Miyazaki et al. (2022) explored how undergraduate design students collaborated with generative AI systems. Their findings revealed that successful collaboration required students to develop new metacognitive strategies for negotiating creative agency with AI systems. Students who viewed the AI as a collaborative partner rather than a tool reported more satisfying creative experiences and produced more innovative work. This suggests that the framing of human-AI relationships significantly influences educational outcomes in co-creative contexts.

In language education specifically, Williams and Patel (2023) examined how ESL students used generative AI for writing development. They found that AI co-creation facilitated language acquisition by providing immediate feedback and modeling, while also challenging students to articulate their ideas more precisely through prompt refinement. This dual benefit—enhancing both language skills and computational thinking—highlights the potential educational value of the “Writing is Coding” approach.

However, research has also identified challenges in implementing AI co-creation in educational settings. Ethical concerns regarding authorship, originality, and assessment have been raised by several scholars (Garcia & Thompson, 2024; Lee, 2023). Additionally, studies have noted significant variations in how students engage with AI systems, with some struggling to develop effective collaboration strategies (Kwon et al., 2023). These challenges underscore the need for structured pedagogical frameworks that support students in developing both the technical and conceptual skills required for meaningful AI co-creation.

2.3 Multimodal Learning and Digital Literacy

The expansion of AI capabilities across text, image, and video modalities has significant implications for multimodal learning and digital literacy. Traditional approaches to multimodal composition have emphasized how different semiotic resources (text, image, sound; etc.) combine to create meaning (Kress & van Leeuwen, 2021). With generative AI, students can now move fluidly between modalities through natural language prompts, creating new possibilities for multimodal expression and learning.

Research by Tan and Lim (2023) examined how secondary school students engaged with multimodal AI tools in arts education. They found that students developed a more sophisticated understanding of modal affordances when they could easily translate concepts between textual, visual, and auditory forms. This “transmodal fluency” enabled students to identify which aspects of their creative vision were best expressed in each modality, leading to more nuanced multimodal compositions.

Similarly, Hernandez-Leo and Asensio-Pérez (2023) investigated how university students used AI co-creation tools in project-based learning. Their findings indicated that multimodal AI collaboration helped students develop more comprehensive digital literacy skills, including critical evaluation of AI-generated content, strategic planning across modalities, and ethical consideration of AI use. These competencies align with expanded definitions of digital literacy that encompass not only technical skills but also critical, creative, and ethical dimensions (Pangrazio & Sefton-Green, 2021).

The progression from text to image to video creation represents increasing complexity in both technical requirements and semiotic resources. As Jewitt and Kress (2023) argue, each modality offers distinct affordances for meaning-making, requiring students to develop specific literacies for effective expression. In AI co-creation contexts, this progression also involves increasing complexity in prompt engineering, as visual and temporal elements introduce new variables that must be articulated through language. Understanding how students navigate this progression is crucial for developing pedagogical approaches that support multimodal literacy development in AI-enhanced learning environments.

2.4 Grounded Theory in AI-Human Collaboration Research

Grounded theory methodology offers a particularly valuable approach for investigating emerging phenomena such as AI-human co-creation, where theoretical frameworks are still evolving. Developed by Glaser and Strauss (1967), grounded theory prioritizes the emergence of theoretical insights from data rather than testing predetermined hypotheses. This inductive approach is especially appropriate for exploring how students engage with novel technologies and develop new literacy practices.

Thornberg and Dunne (2019) highlight the value of grounded theory in creative research contexts, noting that it allows researchers to capture the complex, dynamic processes through which individuals make meaning of their experiences. In the context of AI co-creation, grounded theory enables researchers to identify patterns in how students conceptualize, negotiate, and adapt to collaborative relationships with AI systems without imposing preconceived frameworks that might not adequately capture these novel interactions.

The methodological flexibility of grounded theory also accommodates the multimodal nature of AI co-creation. As Charmaz (2014) argues, grounded theory can incorporate diverse data sources—including visual materials, process documentation, and reflective accounts—making it well-suited to research that spans textual, visual, and temporal modalities. This flexibility allows researchers to trace how students' engagement with AI systems evolves across different creative contexts and expressive forms.

Recent adaptations of grounded theory have specifically addressed technological contexts. Nelson (2020) introduces Computational Grounded Theory (CGT), which integrates computational methods with traditional qualitative analysis to handle the scale and complexity of digital data. Similarly, Lin et al. (2025) propose AI-assisted qualitative analysis techniques that incorporate text mining and pattern recognition while maintaining the interpretive depth central to grounded theory. These methodological innovations offer promising approaches for analyzing the rich, multimodal data generated through AI-human co-creation.

In educational research specifically, grounded theory has proven valuable for understanding how new technologies reshape learning processes and literacy practices. Lassig (2022) demonstrates how grounded theory can illuminate the complex interplay between technological affordances, pedagogical approaches, and student experiences in digital learning environments. This methodological tradition informs our approach to investigating how the “Writing is Coding” paradigm manifests in students' engagement with AI co-creation across multiple modalities.

2.5 Theoretical Framework for AI-Human Collaboration

Several theoretical frameworks have been proposed to conceptualize AI-human collaboration in creative and educational contexts. Distributed cognition theory (Hutchins, 1995; Clark, 2008) offers one valuable perspective, suggesting that cognitive processes extend beyond individual minds to

encompass interactions with tools, technologies, and other people. In this view, AI co-creation can be understood as a form of extended cognition, where creative thinking is distributed across human and computational agents.

Building on this foundation, Wang and Prather (2024) proposed the “Collaborative Intelligence Framework” specifically for AI-human creative partnerships. This framework identifies four key dimensions of collaboration: initiative (who directs the creative process), contribution (what each agent provides), negotiation (how conflicts are resolved), and attribution (how credit is assigned). Their research with creative professionals suggests that successful collaboration requires balance across these dimensions, with both human and AI agents contributing meaningfully to the creative process.

From an educational perspective, Vygotsky's (1978) concept of the zone of proximal development has been adapted to understand AI-supported learning. Ramirez and Chen (2023) argue that AI co-creation tools can function as “dynamic scaffolds” that adapt to students' developing capabilities, providing support that gradually fades as students gain proficiency. This perspective helps explain how AI collaboration might facilitate learning across increasingly complex multimodal tasks.

Critical perspectives on AI-human collaboration have also emerged in recent scholarship. Drawing on posthumanist theory, Hayles (2021) examines how AI co-creation challenges traditional humanist notions of authorship, creativity, and agency. She argues that these technologies require us to reconceptualize creativity as an emergent property of human-machine assemblages rather than an exclusively human capacity. Similarly, Crawford (2023) highlights how AI systems embed and reproduce social biases, raising important questions about power dynamics in AI-human collaboration.

The “Writing is Coding” framework integrates these theoretical perspectives by positioning natural language as the interface between human intention and computational execution. This approach acknowledges the distributed nature of cognition in AI co-creation while maintaining a focus on human agency through strategic language use. By conceptualizing writing as a form of programming, this framework offers a way to understand how humans can maintain creative direction while leveraging the generative capabilities of AI systems across multiple modalities.

3. METHODOLOGY

3.1 Research Design

This study employed a qualitative research design grounded in constructivist epistemology to investigate how students engage with AI co-creation across multiple modalities. The researchers designed a three-round experimental approach that progressively introduced participants to increasingly complex forms of AI collaboration: text generation in Round 1, image creation in Round 2, and video production in Round 3. This sequential design allowed us to observe how students' understanding of the “Writing is Coding” paradigm evolved as they navigated different modalities.

The research was conducted over a 16-week semester, with each round lasting approximately five weeks. Data collection methods included student reflection journals, teacher observation logs, and semi-structured interviews. This triangulation approach provided multiple perspectives on the AI co-creation process and enhanced the credibility of our findings (Creswell & Poth, 2018).

The researchers adopted a grounded theory methodology (Charmaz, 2014; Thornberg & Dunne, 2019; Lin et al., 2025) to analyze the collected data, allowing theoretical insights to emerge inductively from participants' experiences rather than testing predetermined hypotheses. This approach was particularly appropriate given the exploratory nature of our research questions and the rapidly evolving landscape of AI co-creation in educational contexts.

The study addressed potential bias through several methodological controls. First, multiple data sources were triangulated to ensure comprehensive understanding of student experiences. Second, the research team included members with diverse disciplinary backgrounds to provide multiple analytical perspectives. Third, member checking was conducted with a subset of participants to validate emerging

themes and interpretations. Fourth, an external auditor reviewed the coding process and theoretical development to ensure methodological rigor.

The study employed purposive sampling to recruit 102 undergraduate students from diverse academic disciplines at a large public university. This sampling method was selected to ensure representation across different fields of study and varying levels of prior experience with AI tools. While purposive sampling may limit generalizability, it was deemed appropriate for this exploratory study because it allowed the researchers to capture diverse perspectives on AI co-creation across disciplinary boundaries. The goal was not statistical generalization but rather theoretical understanding of how students from different academic backgrounds engage with the “Writing is Coding” paradigm.

3.2 Participants

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Participants ranged in age from 18 to 24 years ($M = 20.3$, $SD = 1.7$) and represented 15 different academic majors, including humanities (32%), social sciences (28%), STEM fields (25%), and arts (15%). Prior experience with AI tools varied significantly: 34% reported no previous experience, 41% had limited experience with text-based AI systems, and 25% had some experience with multimodal AI tools. This diversity in backgrounds and experience levels provided rich variation for theoretical sampling and analysis.

Participants were selected using purposive sampling to ensure representation across different levels of prior experience with AI tools. Based on a preliminary survey, 28% of participants reported extensive prior experience with AI tools, 45% reported moderate experience, and 27% reported minimal or no prior experience. This distribution allowed us to examine how prior experience influenced engagement with the “Writing is Coding” framework.

All participants provided informed consent, and the study was approved by the university's Institutional Review Board. To protect participant privacy, all data were anonymized, and pseudonyms are used throughout this paper.

Ethical approval was obtained from the university's Institutional Review Board, and all participants provided informed consent. Students were assured that participation was voluntary and would not affect their academic standing. Data collection procedures followed strict confidentiality protocols, with all identifying information removed during the analysis phase.

3.3 Experimental Procedures

3.3.1 Round 1: Text-Based AI Co-Creation

The first round focused on text-based AI co-creation, introducing students to the fundamental principles of the “Writing is Coding” framework. Participants were tasked with collaboratively writing a short narrative with an AI language model. The process involved three phases:

1. Orientation Phase: Students received instruction on prompt engineering principles and participated in workshops demonstrating effective strategies for AI collaboration. They were introduced to the concept of “Writing is Coding” and encouraged to think of their prompts as instructions that guide the AI's output.

2. **Creation Phase:** Working individually, students developed a narrative concept and collaborated with the AI system to produce a 1000-1500 word story. They were required to document their prompt strategies, the AI's responses, and their iterative refinements in their reflection journals.
3. **Reflection Phase:** Students participated in small group discussions to share their experiences and insights. They then completed a structured reflection addressing questions about their prompt strategies, challenges encountered, and perceptions of the human-AI collaborative relationship.

Throughout this round, instructors maintained observation logs documenting students' engagement, questions, and apparent challenges. These observations provided valuable contextual information to complement students' self-reported experiences.

3.3.2 Round 2: Image-Based AI Co-Creation

The second round expanded into visual modality, challenging students to create images through AI collaboration. Building on their text-based experience, participants now needed to translate visual concepts into textual prompts—a process requiring different cognitive and descriptive skills. The round followed a similar three-phase structure:

1. **Orientation Phase:** Students received instruction on visual prompt engineering, including techniques for describing style, composition, lighting, and emotional tone. Examples of effective and ineffective visual prompts were analyzed and discussed.
2. **Creation Phase:** Participants created a series of five thematically related images using AI image generation tools. They were encouraged to experiment with different prompt strategies and to document how specific language choices influenced visual outcomes.
3. **Reflection Phase:** Students analyzed their most and least successful images, identifying patterns in prompt effectiveness. They completed reflection journals addressing how the transition from text to image generation affected their understanding of the “Writing is Coding” paradigm.

During this round, particular attention was paid to how students translated abstract concepts and emotions into concrete visual descriptions. Teacher observation logs focused on students' adaptive strategies as they navigated the challenges of visual prompt engineering.

3.3.3 Round 3: Video-Based AI Co-Creation

The final round introduced the most complex modality: video creation through AI collaboration. This phase required students to consider not only visual elements but also temporal progression, movement, and narrative coherence across time. The round maintained the established structure:

1. **Orientation Phase:** Students learned about video prompt engineering, including techniques for describing motion, scene transitions, narrative progression, and temporal elements. They analyzed examples of AI-generated videos and the prompts that produced them.
2. **Creation Phase:** Participants created short (30-60 second) videos using AI video generation tools. They were encouraged to build upon concepts explored in previous rounds, potentially extending their text narratives or image series into the video format.
3. **Reflection Phase:** Students completed comprehensive reflections on their three-round journey, addressing how their understanding of “Writing is Coding” evolved across modalities. They identified transferable strategies and modality-specific challenges.

Semi-structured interviews were conducted with a subset of 30 participants after the completion of all three rounds. These interviews explored participants' holistic experience across the three

modalities and their evolving understanding of the relationship between writing and coding in AI co-creation contexts.

Figure 1. Image and Video generated in the experiment process.



Figure 2. Image and Video generated in the experiment process.



Figure 3. Image and Video generated in the experiment process.



Figure 4. Image and Video generated in the experiment process.



Figure 5. Image and Video generated in the experiment process.



3.3.4 Data Analysis

Data analysis followed the constructivist grounded theory approach outlined by Charmaz (2014) and Thornberg and Dunne (2019), and AI-assisted grounded theory approach by Lin et al., (2025), involving three main coding phases: initial coding, focused coding, and theoretical coding. This process was applied to all data sources (reflection journals, observation logs, and interview transcripts) to develop a comprehensive understanding of participants' experiences.

3.3.4.1 Initial Coding

During initial coding, we analyzed data line-by-line to identify significant processes, actions, and perspectives. This phase employed in vivo coding (using participants' own language) and process coding (focusing on actions) to stay close to the data. Two researchers independently coded a subset of the data (20%) to establish coding consistency, achieving an inter-rater reliability of 87% before proceeding with the full dataset.

3.3.4.2 Focused Coding

In the focused coding phase, we identified the most significant and frequent initial codes and used them to categorize the data more selectively. This process involved comparing data to data, data to codes, and codes to codes to develop more conceptual categories. Through this comparative analysis, we identified recurring patterns and emerging themes across participants' experiences.

3.3.4.3 Theoretical Coding

The final phase involved theoretical coding to identify relationships between the focused codes and integrate them into a coherent analytical framework. This process was guided by constant comparison and theoretical memoing, allowing us to develop a conceptual understanding of how students engaged with the “Writing is Coding” paradigm across different modalities.

Throughout the analysis, we employed several strategies to enhance trustworthiness, including triangulation of data sources, member checking with participants, peer debriefing among researchers, and maintaining an audit trail of analytical decisions. These measures helped ensure that our findings accurately represented participants' experiences while acknowledging the interpretive nature of qualitative analysis.

The data analysis process was iterative and recursive, with ongoing refinement of codes and categories as new data were collected and analyzed. The use of AI-assisted analysis tools helped manage the large volume of textual data while maintaining the interpretive depth required for grounded theory methodology.

4. RESULTS

The analysis revealed five interconnected themes that characterize student engagement with the “Writing is Coding” paradigm across multimodal AI co-creation. These themes emerged consistently across all three rounds and were evident in multiple data sources. Figure 1 through Figure 5 illustrate key patterns in student engagement, while Figure 6 provides a comprehensive overview of the thematic framework.

4.1 Prompt Engineering as Algorithmic Thinking

Across all three rounds, students progressively developed an algorithmic mindset when crafting prompts for AI systems. This cognitive shift was particularly evident in how students approached the iterative refinement of their instructions to achieve desired outcomes.

In Round 1 (text creation), most students initially approached prompts as simple requests rather than structured instructions. Their early journal entries revealed a tendency to use vague, general language that produced inconsistent results. For example, one student wrote: “At first I just asked for 'a

story about friendship' and got something generic. I didn't realize I needed to be specific about tone, setting, and characters” (Student 37, Reflection Journal).

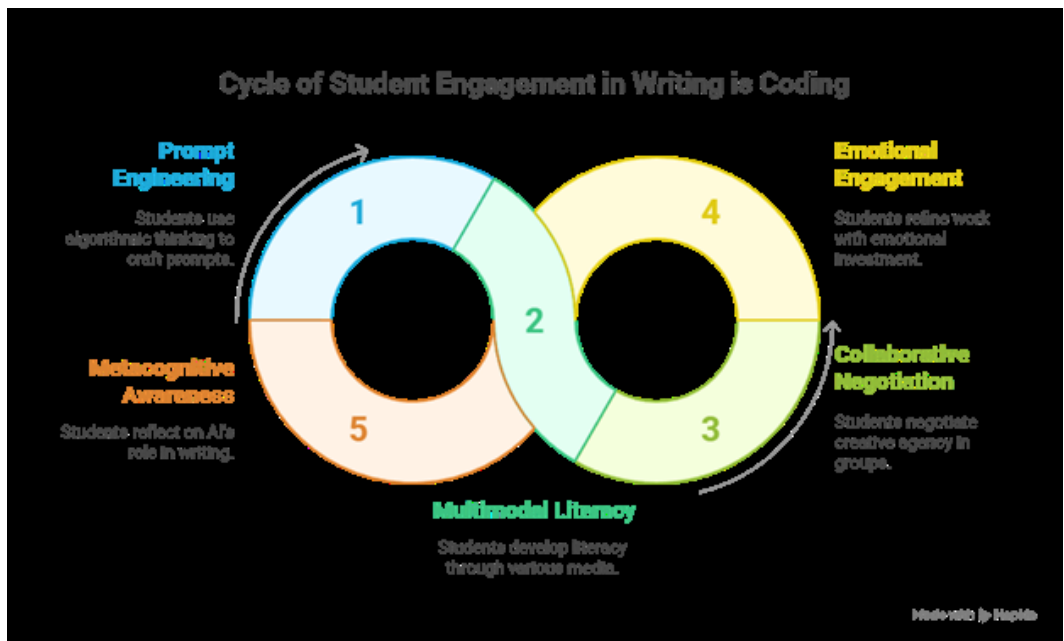
As students progressed through the rounds, they began to adopt more systematic approaches to prompt construction. By Round 2 (image creation), 78% of participants reported using structured prompt templates that specified multiple parameters such as style, composition, lighting, and mood. Teacher observation logs noted that students frequently used conditional statements in their prompts (e.g., “If the foreground shows X, then the background should show Y”), demonstrating an emerging understanding of algorithmic logic.

The most sophisticated algorithmic thinking emerged in Round 3 (video creation), where students needed to account for temporal progression and movement. Interview data revealed that students consciously approached video prompts as “mini-programs” with sequential instructions:

“I realized I needed to think like a programmer-breaking down the video into scenes, defining transitions, and specifying how elements should change over time. It wasn't just describing what I wanted to see, but programming how the sequence should unfold.” (Student 85, Journal)

This evolution toward algorithmic thinking was reflected in prompt complexity across the three rounds. Analysis of students' documented prompts showed an increase in average word count (Round 1: 42 words; Round 2: 87 words; Round 3: 124 words) and the use of structured formatting elements such as numbered sequences, conditional statements, and parameter specifications, as shown in figure 6.

Figure 6. Five Themes of Student Engagement in Writing is Coding



4.2 Multimodal Literacy Development

The progression from text to image to video creation facilitated significant development in students' multimodal literacy-their ability to understand and leverage the affordances of different communicative modes.

In Round 1, students primarily focused on narrative elements and linguistic expression. Their reflections demonstrated awareness of how word choice, tone, and structure influenced AI-generated

text. However, when transitioning to image creation in Round 2, many students initially struggled to translate their ideas into visual language:

“I knew exactly what I wanted in my mind, but I couldn't find the right words to describe it visually. I had to learn a whole new vocabulary for describing composition, perspective, and visual style.” (Student 14, Journal)

Teacher observation logs noted that students frequently consulted online resources for visual terminology and studied examples of effective image prompts. By the middle of Round 2, many had developed specialized vocabularies for visual description, with 63% of students reporting increased confidence in their ability to “translate” between textual and visual modes.

The transition to video creation in Round 3 required further expansion of multimodal literacy to include temporal elements. Students needed to develop language for describing movement, pacing, transitions, and narrative progression across time. Analysis of reflection journals showed that students increasingly recognized the distinct affordances of each modality:

“Text is good for internal thoughts and abstract concepts. Images capture a moment and emotion. But video adds the dimension of change and movement—it shows relationships between moments. Each requires different instructions for the AI.” (Student 72, Reflection Journal)

By the end of the three rounds, 89% of participants demonstrated what we term “transmodal fluency”—the ability to strategically select and combine modalities to express different aspects of their creative vision. This development was particularly evident in how students approached complex concepts that required multiple modalities for full expression.

4.3 Collaborative Negotiation of Creative Agency

A central theme across all rounds was how students negotiated creative agency with AI systems. This negotiation evolved from initial expectations of complete control to more nuanced collaborative relationships.

In Round 1, many students approached the AI as a tool that should precisely execute their instructions. When the system produced unexpected outputs, students often expressed frustration and framed these deviations as errors rather than creative contributions:

“It kept adding details I didn't ask for and taking the story in directions I didn't intend. I had to keep correcting it to follow my original plan.” (Student 43, Journal)

As students progressed to image creation in Round 2, their approach to agency began to shift. Teacher observations noted that students increasingly welcomed unexpected elements in AI-generated images, often incorporating these “happy accidents” into their creative process. By Round 3, many students had adopted what one participant described as “collaborative improvisation”:

“I've stopped trying to control every detail. Now I give a strong direction but leave room for the AI to surprise me. It's like jazz—I establish the key and tempo, but we improvise together.” (Student 29, Journal)

This evolution was reflected in prompt strategies, which shifted from highly prescriptive instructions in early rounds to more balanced approaches that combined specific requirements with space for AI contribution. Analysis of reflection journals revealed that by the final round, 72% of students described their relationship with the AI using collaborative language (e.g., “partner,” “collaborator,” “co-creator”) rather than instrumental terms (e.g., “tool,” “assistant”).

However, this collaborative approach was not universal. Approximately 18% of participants maintained a strong preference for directive control throughout all three rounds, viewing any unexpected AI contributions as deficiencies rather than creative additions. These students tended to frame the “Writing is Coding” paradigm in more technical terms, emphasizing precision and predictability over creative exploration.

4.4 Emotional Engagement Through Iterative Refinement

An unexpected finding was the significant emotional dimension of AI co-creation, particularly how the iterative refinement process engaged students' emotional investment in their work.

In Round 1, students primarily described their emotional responses in terms of satisfaction or frustration with outputs. However, as they progressed to visual modalities in Rounds 2 and 3, their reflections revealed deeper emotional engagement with both the process and the products of co-creation:

“When the image finally captured exactly what I had imagined—this feeling of nostalgic summer twilight—I actually got emotional. It was like the AI had reached into my memory and made it visible.” (Student 61, Reflection Journal)

Teacher observation logs noted increased emotional investment during the iterative refinement process, with students showing visible excitement, disappointment, or surprise as they received AI outputs. This emotional engagement appeared to intensify as students moved from text to image to video creation, with the highest emotional responses observed during video generation.

Interview data suggested that this emotional dimension was connected to the externalization of internal concepts—seeing one's ideas manifested in concrete form. As one student explained:

“There's something powerful about watching your words transform into something visual. It's like the AI is making your imagination tangible, and that creates a strong emotional connection to the work.” (Student 8, Journal)

This emotional engagement appeared to motivate more extensive iteration. Analysis of process documentation showed that students conducted an average of 3.2 iterations per text creation, 5.7 iterations per image, and 7.4 iterations per video. This increasing investment of time and effort suggests that emotional connection to the creative process deepened as students progressed through the modalities.

4.5 Metacognitive Awareness of Human-AI Boundaries

The final theme that emerged across all three rounds was students' developing metacognitive awareness of the boundaries between human and AI contributions to the creative process.

In early reflections, students often described a binary view of authorship, questioning whether they or the AI should be considered the “real” creator of the outputs. As they progressed through the rounds, their reflections demonstrated increasingly nuanced understanding of distributed creativity:

“I've realized there's no clear line between my contribution and the AI's. My words shape its output, but its interpretation shapes my next prompt. It's a continuous feedback loop where authorship is shared.” (Student 93, Journal)

This metacognitive awareness extended to recognition of the AI's limitations and biases. By Round 3, 84% of students demonstrated critical awareness of how AI systems reflected patterns from their training data, often identifying when outputs reproduced common tropes or stereotypes. This awareness prompted many students to deliberately push against these tendencies in their prompts:

“I noticed the AI kept generating conventional Hollywood-style scenes even when I asked for something experimental. I had to specifically counter those tendencies by referencing alternative film traditions and explicitly rejecting mainstream aesthetics.” (Student 17, Journal)

Perhaps most significantly, students developed awareness of how the “Writing is Coding” paradigm influenced their own creative processes. Many reported that thinking of writing as a form of programming changed not only how they interacted with AI but also how they approached creative expression more generally:

“I find myself thinking more systematically about my creative writing now—breaking down concepts into components, considering logical relationships between elements, and being more precise with language. The coding mindset has carried over into my regular writing.” (Student 52, Journal)

This transfer effect was reported by 67% of participants, suggesting that the “Writing is Coding” framework has potential implications beyond AI co-creation contexts.

5. DISCUSSION

5.1 Theoretical Implications

The evolution of students' prompt engineering strategies toward algorithmic thinking supports the conceptual foundation of the “Writing is Coding” paradigm. As students progressed through increasingly complex modalities, their approach to language became more structured, systematic, and procedural-mirroring key aspects of computational thinking. This alignment between natural language instruction and algorithmic logic suggests that the boundaries between linguistic and computational literacy may be more permeable than traditionally assumed.

The findings extend previous research on computational thinking in non-programming contexts (Vee, 2017; Hammond & Neeman, 2022) by demonstrating how AI co-creation can serve as a bridge between natural language expression and computational processes. The progression from simple requests to structured, parameter-based instructions illustrates how students naturally adopted computational principles when faced with the challenge of guiding AI systems through language.

The development of “transmodal fluency” among participants contributes to theoretical understandings of multimodal literacy in digital environments. While previous research has examined how students navigate between different modalities in traditional composition (Kress & van Leeuwen, 2021; Jewitt & Kress, 2023), The findings highlight the unique challenges and opportunities that arise when natural language serves as the interface for generating content across modalities. The concept of transmodal fluency—the ability to strategically select and combine modalities through language-based instructions—represents a novel dimension of digital literacy specific to AI co-creation contexts.

The negotiation of creative agency between students and AI systems offers insights into distributed creativity theory. Our findings align with Wang and Prather's (2024) Collaborative Intelligence Framework, particularly regarding the dimensions of initiative and contribution. However, the results suggest that this negotiation is not static but evolves as students gain experience across different modalities. The shift from directive control to collaborative improvisation indicates that human-AI creative relationships are dynamic and context-dependent, challenging simplistic notions of AI tools as either autonomous creators or passive instruments.

5.2 Practical Implications

The practical implications of our findings are particularly relevant for educators seeking to integrate AI co-creation into curriculum design. The three-round progression from text to image to video provides a structured approach for introducing students to increasingly complex forms of AI collaboration. This gradual progression allows students to develop foundational skills in prompt engineering before tackling more challenging multimodal tasks.

The results suggest several specific strategies for supporting effective AI co-creation in educational settings:

1. Explicit instruction in prompt engineering: Students benefit from structured guidance in crafting effective prompts, including examples of well-structured instructions and opportunities to analyze the relationship between prompt characteristics and AI outputs.
2. Scaffolded multimodal progression: The sequence from text to image to video creation provides natural scaffolding, allowing students to transfer skills from familiar to unfamiliar modalities while developing increasingly sophisticated prompt strategies.
3. Reflection-focused documentation: Regular reflection on the co-creation process helps students develop metacognitive awareness of their strategies, challenges, and evolving relationship with AI systems.
4. Collaborative analysis of unexpected outputs: When AI systems produce unexpected results, treating these as opportunities for critical analysis rather than mere errors can foster deeper understanding of both AI capabilities and students' own communicative intentions.

5. Explicit discussion of agency and authorship: Creating space for students to explore questions of creative ownership and distributed agency helps them develop a more nuanced understanding of human-AI collaboration.

These practical implications address the challenges identified in previous research regarding the implementation of AI co-creation in educational settings (Garcia & Thompson, 2024; Kwon et al., 2023). By providing structured support for developing both technical skills and conceptual understanding, educators can help students engage meaningfully with AI systems as creative partners.

5.3 Educational Framework Development

Based on our findings, we propose an educational framework for implementing the “Writing is Coding” approach in diverse learning contexts. This framework integrates the five themes identified in our analysis into a coherent pedagogical model that can be adapted across disciplines and educational levels.

The AI Co-Creation Literacy Framework consists of five interconnected dimensions:

1. Algorithmic Expression: Developing the ability to structure natural language as computational instructions, including skills in parameter specification, conditional logic, and sequential organization.
2. Multimodal Translation: Building capacity to move fluidly between textual, visual, and temporal modes through strategic language use, recognizing the unique affordances and constraints of each modality.
3. Collaborative Negotiation: Cultivating strategies for effective human-AI collaboration, including balancing directive control with creative openness and critically evaluating AI contributions.
4. Iterative Refinement: Developing systematic approaches to the revision process, including strategies for analyzing outputs, identifying areas for improvement, and progressively refining prompts.
5. Critical Meta-Awareness: Fostering critical reflection on the nature of human-AI creativity, including questions of authorship, bias, ethical considerations, and the relationship between language and computation.

This framework can be implemented through a spiral curriculum approach, with students revisiting each dimension at increasing levels of complexity as they progress through different modalities and creative tasks. The framework is designed to be flexible, allowing educators to emphasize different dimensions based on specific learning objectives and disciplinary contexts.

Importantly, this framework positions AI co-creation not merely as a technical skill but as a form of critical digital literacy that encompasses creative, analytical, and ethical dimensions. By framing writing as a form of coding, educators can help students develop transferable skills that bridge traditionally separated domains of linguistic and computational thinking.

5.4 Limitations and Future Research

While our study provides valuable insights into how students engage with AI co-creation across modalities, several limitations should be acknowledged. First, the research was conducted with undergraduate students at a single institution, limiting the generalizability of our findings to other educational contexts and age groups. Future research should explore how the “Writing is Coding” framework might be adapted for different educational levels, from K-12 to graduate education. The purposive sampling method, while appropriate for exploratory research, means that findings may not be representative of all student populations.

Second, our study focused on a specific set of AI tools available at the time of research. Given the rapid evolution of generative AI technologies, future studies should examine how emerging capabilities might influence the dynamics of human-AI co-creation and the applicability of the “Writing is Coding” paradigm.

Third, while our qualitative approach provided rich insights into students' experiences and perceptions, it did not include quantitative measures of learning outcomes or creative quality. Future research could employ mixed-methods approaches to examine the relationship between AI co-creation processes and measurable outcomes in terms of both learning and creative production.

Finally, our study was conducted over a single semester, limiting our ability to observe long-term impacts of the “Writing is Coding” approach on students' creative and computational thinking. Longitudinal studies would be valuable for understanding how engagement with AI co-creation might influence students' approaches to both writing and programming over extended periods.

Despite these limitations, our findings provide a foundation for future research on AI co-creation in educational contexts. Particularly promising directions include exploring how the “Writing is Coding” framework might be applied in specific disciplinary contexts, investigating potential transfer effects between AI co-creation and traditional programming education, and examining how different pedagogical approaches might address the ethical and critical dimensions of human-AI collaboration.

6. CONCLUSION

6.1 Summary of Findings

Our analysis revealed five core themes that characterize students' engagement with the “Writing is Coding” paradigm: prompt engineering as algorithmic thinking, multimodal literacy development, collaborative negotiation of creative agency, emotional engagement through iterative refinement, and metacognitive awareness of human-AI boundaries.

As students progressed from text to image to video creation, they developed increasingly sophisticated approaches to prompt engineering, adopting algorithmic thinking strategies such as parameter specification, conditional logic, and sequential organization. This evolution demonstrates how the “Writing is Coding” framework naturally encourages computational thinking through natural language interaction.

The progression across modalities facilitated significant development in students' multimodal literacy, particularly what we term “transmodal fluency”—the ability to strategically select and combine modalities through language-based instructions. This capacity represents an important dimension of digital literacy in an era where AI systems increasingly mediate multimodal creation.

Students' negotiation of creative agency with AI systems evolved from initial expectations of complete control to more nuanced collaborative relationships, with most participants ultimately adopting a balanced approach that combined specific direction with openness to AI contributions. This evolution reflects the complex nature of human-AI co-creation, where creativity emerges through dynamic interaction rather than unilateral control.

The emotional dimension of AI co-creation proved particularly significant, with students demonstrating increasing emotional investment as they progressed through the modalities. This emotional engagement appeared to motivate more extensive iteration and refinement, suggesting that AI co-creation can foster a deep personal connection to creative work despite the technological mediation.

Finally, students developed metacognitive awareness of the boundaries between human and AI contributions, recognizing both the distributed nature of creativity in AI collaboration and the influence of the “Writing is Coding” paradigm on their own creative processes. This awareness extended beyond the experimental context, with many students reporting transfer effects to their broader understanding of both writing and computational thinking.

6.2 Contributions to the Field

This research makes several significant contributions to our understanding of AI co-creation in educational contexts. First, it provides empirical support for the conceptual framework of “Writing is Coding,” demonstrating how this paradigm manifests across different modalities and stages of the creative process. The alignment between students' evolving prompt strategies and computational thinking principles suggests that AI co-creation can serve as a bridge between linguistic and computational literacy.

Second, the study offers a structured approach for introducing students to increasingly complex forms of AI collaboration through the three-round progression from text to image to video. This approach provides natural scaffolding for developing both technical skills and conceptual understanding, addressing challenges identified in previous research on implementing AI co-creation in educational settings.

Third, the proposed AI Co-Creation Literacy Framework integrates the five themes identified in our analysis into a coherent pedagogical model that can be adapted across disciplines and educational levels. This framework positions AI co-creation not merely as a technical skill but as a form of critical digital literacy that encompasses creative, analytical, and ethical dimensions.

Finally, the study contributes to theoretical discussions about the changing nature of authorship, creativity, and agency in AI-mediated contexts. The findings challenge binary distinctions between human and machine creativity, suggesting instead that meaningful co-creation emerges through dynamic negotiation and collaborative improvisation.

6.3 Addressing the Research Questions

Returning to our initial research questions, this study provides several key insights:

Regarding how students conceptualize and adapt their writing processes when engaging in AI co-creation across modalities (Question 1), we found that students progressively reconceptualized writing as a form of computational instruction rather than merely expressive communication. Their adaptation strategies evolved from simple descriptive prompts to structured, parameter-based instructions that demonstrated increasing algorithmic sophistication. This evolution was particularly evident in the transition from text to visual modalities, where students developed specialized vocabularies and structural approaches to translate abstract concepts into concrete visual parameters.

In terms of cognitive and creative strategies that emerge through multimodal AI collaboration (Question 2), we identified several key patterns: strategic iteration based on systematic analysis of outputs; collaborative negotiation that balanced directive control with openness to AI contributions; modal translation strategies that leveraged the unique affordances of each modality; and metacognitive reflection on the relationship between language choices and AI outputs. These strategies became more sophisticated as students progressed through increasingly complex modalities, with the most advanced approaches emerging during video creation.

Finally, regarding how the “Writing is Coding” framework influences students' understanding of computational processes and creative expression (Question 3), our findings suggest that this conceptual framework facilitated deeper engagement with both domains. Students developed more sophisticated computational thinking skills—including abstraction, pattern recognition, and algorithmic logic-through natural language interaction rather than conventional programming. Simultaneously, they gained new perspectives on creative expression, recognizing how structured language could shape multimodal outputs while maintaining personal voice and aesthetic vision. This integration of computational and creative thinking represents a promising direction for education in an increasingly AI-mediated world.

6.4 Final Thoughts

As generative AI technologies become increasingly integrated into educational and creative contexts, the relationship between natural language and computational processes takes on new

significance. The “Writing is Coding” paradigm offers a valuable conceptual framework for understanding and navigating this relationship, positioning writing not merely as expression but as a form of programming that directs AI systems across multiple modalities.

Our findings suggest that this paradigm has significant potential for enhancing both creative expression and computational thinking in educational settings. By engaging with AI co-creation across modalities, students develop not only technical skills in prompt engineering but also deeper understanding of how language shapes computational outputs and how creativity emerges through human-machine collaboration.

As we continue to explore the educational possibilities of AI co-creation, the “Writing is Coding” framework reminds us that the boundaries between traditionally separated domains-between humanities and computer science, between expression and computation, between human and machine creativity-are increasingly permeable. By embracing this permeability rather than reinforcing traditional divisions, we can help students develop the integrated literacies they need to navigate and shape an AI-augmented future.

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